Hierarchical Cancer Classification Report

AI-assisted decision support summary for brain_tumor_0002.jpg

Executive Summary

Date & Time	2026-04-19 15:50:51	**Device**	cpu
Organ Model	Ready (resnet50_organ_classifier.pth)	**Subtype Model**	Ready (resnet50_subtype_classifier_best.pth)

Final Decision

Pituitary Tumor

Outcome: ABNORMAL Confidence: 97.01%

Label: Pituitary Tumor Pipeline Status: HIGH_CONFIDENCE

Tissue Route: Brain Tissue Subtype / Outcome Detail: Pituitary Tumor

AI-generated decision support summary prepared for review.

Visual Review

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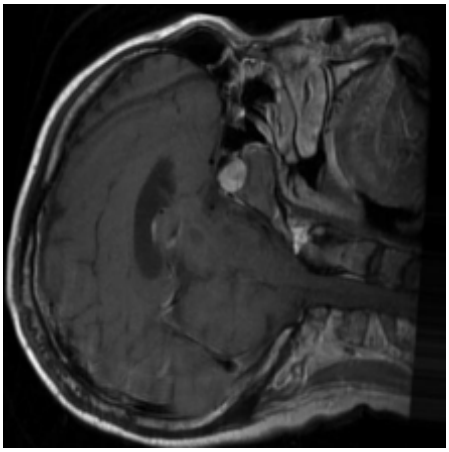
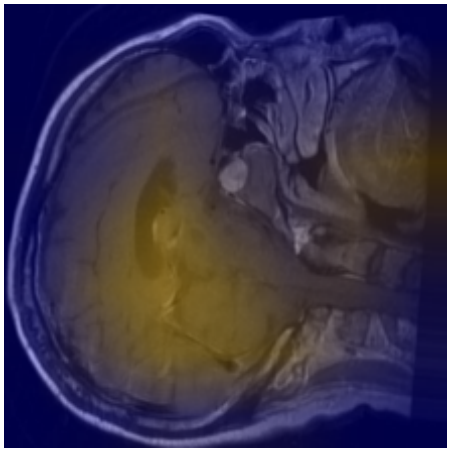
Input Scan



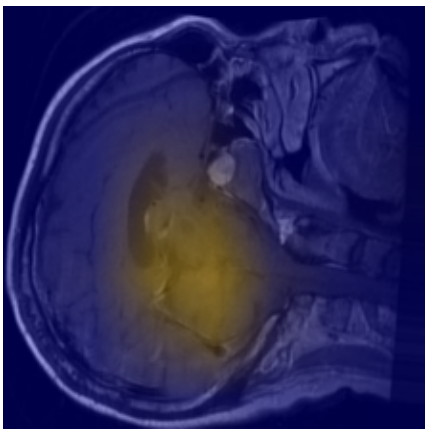
Submitted image used for analysis.

Grad-CAM: Organ Routing



Brain Tissue

Grad-CAM: Final Outcome



Pituitary Tumor

Clinical Routing Summary

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Outcome	ABNORMAL
Confidence	97.01%
Entropy	0.16
Label	Healthy Brain Tissue
Reason	None

Organ / Tissue Probabilities

Organ	Probability (%)
Brain Tissue	99.71%
Cervical Tissue	0.09%
Kidney Tissue	0.05%
Lung & Colon Tissue	0.04%
Breast Tissue	0.04%
Lymphatic System	0.03%
Oral Tissue	0.03%
Blood (Leukemia-related)	0.01%

Subtype Probabilities

Subtype	Probability (%)
Pituitary Tumor	97.01%
Healthy Brain Tissue	2.17%
Brain Meningioma	0.60%
Brain Glioma	0.21%

Warnings / Notes

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- Relaxed validation applied for MRI.